

From Data to Digital Transformation

Remote Monitoring of Chronic Patients (HealthTech) — Part II Final Report

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Abstract—Part I placed Mexico fourth of five economies on the Digital Readiness Score (DRS = 23.0/100): near-universal 4G coverage, but weak affordability, fixed-broadband depth and R&D. We propose a mobile-first pipeline that turns wearable and smartphone signals into a 48-hour risk score for chronic-care coordinators, sold as B2B SaaS. It is a hypothesis to be validated in a pilot, not a demonstrated result.

I. Problem and Evidence

Problem. Adults with hypertension, type-2 diabetes, COPD or heart failure are observed mainly at scheduled visits; deterioration is often detected only at the emergency department (ED). *Goal:* give the care coordinator of a hospital chronic-care program a daily ranked list of patients with elevated probability of an unplanned ED visit or admission within 48 h, $P(y_{48h} = 1 \mid \mathbf{x})$. *Phenomenon:* drift of SpO₂, resting heart rate, blood pressure and glucose from the patient's own baseline. *Outcome metric:* unplanned ED visits/admissions per 1,000 monitored patient-months.

Evidence. Part I [1]–[3]: 4G coverage 97.4% and 83.1% Internet users (feasible channel); mobile basket 1.49% of GNI per capita, the costliest of the five, and R&D 0.25% of GDP (equity and capability risk). ICT-goods exports (12.45%) reflect assembly, not domestic capacity (DRS without them: 12.0; rank unchanged). Clinical burden: national prevalence of type-2 diabetes is 17.0% (11.6% diagnosed), Ensanut Continua 2021–2024 [4]. This does not measure avoidable ED use.

II. Five Sectors

TABLE I
SECTOR MATRIX: DIGITAL LAYER → DATA → DECISION.

Sector	Digital layer	Data	Decision
Telecom	IoT probes, telemetry	Traffic, latency, loss	Reroute / add capacity
Health	Wearables, EHR	HR, SpO ₂ , glucose, visits	Prioritize follow-up, triage
Agriculture	Drones, soil sensors	Moisture, nutrients, imagery	Precision irrigation
Science	NGS sequencers	DNA/RNA sequences	Personalized therapy
Energy	Smart meters, grid sensors	Hourly load, voltage, outages	Dynamic tariffs, dispatch

III. Pipeline

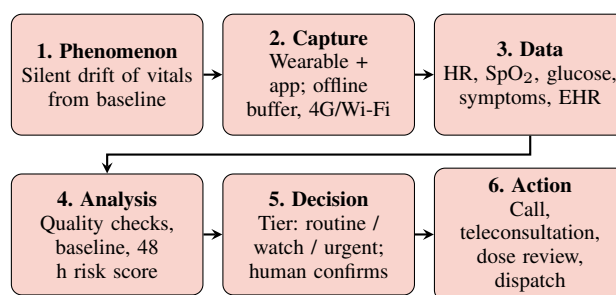


Fig. 1. Transformation pipeline. Confirmed events, false alarms and misses return as labels for retraining.

Example (data → action). SpO₂ every 10 min; after quality filtering it falls 98%→90% in 6 h against a stable personal baseline; with heart rate, symptoms and history the risk crosses the urgent threshold (illustrative, to be personalized and clinically validated). The coordinator replaces next week's visit with same-day teleconsultation; a mobile unit is sent only after clinician confirmation.

IV. Analytical Strategy

TABLE II
FOUR LEVELS OF ANALYSIS FOR THE SAME PROBLEM.

Level	Question, data, technique	Result → decision
Descriptive	How many patients had out-of-range vitals yesterday; ED visits last month? Vitals, ED records; SQL, control charts.	Daily alert board → staffing of review team.
Diagnostic	Why did decompensations rise in one clinic this week? EHR, adherence, weather, device; stratified regression, change-point.	Ranked risk factors → targeted campaign.
Predictive	Who will have an unplanned ED visit/admission in 48 h? Labeled history; logistic regression, gradient boosting, calibration.	Risk score (AUROC, PR-AUC) → proactive contact.
Prescriptive	Given limited callers, slots and ambulances, whom to contact first? Scores, schedules, travel times; integer programming.	Daily contact list → resource allocation.

Not interchangeable: “120 patients are out of range” describes the past, not who deteriorates tomorrow; “patient A has 38% risk” does not say what to do, since action depends on capacity, cost and intervention effectiveness. Only the prescriptive level combines risk with constraints, and it needs pilot evidence that the action reduces events.

V. Business Model, Platform, Network Effects

Value. *Creation:* fewer avoidable ED visits and earlier care for hospitals, payers and patients. *Delivery:* patient app plus low-cost wearable; clinician dashboard integrated with hospital records. *Capture:* monthly fee per active monitored patient plus integration fees, priced in the pilot from the cost of avoided events. **Model:** B2B SaaS with a secondary data-driven component (de-identified analytics, only with consent). Rejected as primary: advertising (trust), marketplace, freemium (follow-up has marginal cost), on-demand. Given the affordability gap, patients should not bear data costs (zero-rated or data-light traffic).

Platform. Partly: it mediates patients, clinicians/hospitals and payers (researchers and device makers as a governed fourth group). Rules: consent and revocation, role-based access, audit logs, interoperability, clinical validation before model updates, SLAs, human liability for the final decision. Interactions generate vitals, alert outcomes, overrides and response times (labels). **Network effects.** Direct effects are expected to be weak: adding more patients does not automatically improve each patient’s service, since timely care still depends on clinical capacity and data quality. Indirect effects could be moderate and regional: more participating hospitals could attract patients and payers, while more enrolled patients could make participation more attractive to hospitals and payers. Data learning: plausible but with diminishing returns and only if data are representative. Negative: alert fatigue when precision is low.

VI. Scalability (10× users)

Cloud ingestion and compute grow sub-linearly to linearly; storage grows linearly (tiering and down-sampling help); network load is borne by users’ data plans; **alert review by staff is the bottleneck** unless precision improves; hospital-by-hospital sales and support persist; regulation adds recurring audits and re-validation. Technically scalable, clinically limited by human review. Slow-growing costs: software, architecture, model design. Continuously growing: compute, storage, support, clinical operations, compliance, devices. Scalability is not zero cost: $c = c_{var} + F/N$; fixed costs dilute but $c_{var} > 0$.

VII. Data Quality and Responsibility

Quality: uncalibrated sensors give false readings → alert fatigue or missed events (quality flags, confirmation before dispatch). **Representativeness:** a model trained on U.K. or Estonian patients may misestimate Mexican risk; enrollment is self-selected toward connected people (local validation, stratified metrics). **Privacy:** health data breaches enable discrimination (minimization, encryption, consent; compliance with Mexican law to be confirmed with counsel). **Bias:** learning from premium-device users under-serves rural, low-income patients (subgroup audits). **Correlation ≠ causation:** heavy app users may show more hospitalizations because they are sicker (confounding) or are monitored more (detection bias). In Part I, broadband and R&D correlate at $r \approx 0.98$ but with $n = 5$ and common drivers this shows no causality.

VIII. Patents and Innovation

A patent is a time-limited exclusive right (typically 20 years) over an invention in exchange for disclosure [8]. *Novelty* means that the invention adds a technical feature not disclosed in the prior art. *Inventive step* means that the proposed solution is not obvious to a person with ordinary skill in the relevant technical field. *Industrial applicability* means that the invention has a practical industrial or business use beyond a theoretical idea [8]. Candidate inventive component: edge sensor-fusion and adaptive sampling that keeps a personalized risk baseline under intermittent connectivity and low-cost sensors; pure math or business methods are often excluded, so it needs a technical effect and IP counsel. IPC class G16H covers healthcare informatics [6]; main area: **computer technology**, secondary digital communication (H04L) and medical technology (A61B) [7]. WIPO consultation (Table III, Fig. 2) uses field-level counts as a proxy for G16H; they are publications at each national office, including foreign filers, so they show office activity, not local inventiveness (Mexico’s office lists more medical-technology publications than the U.K.’s, yet Part I shows far lower resident-patent intensity). Many patents do not prove innovation quality, adoption or success: counts measure filings, and the U.K. has about 20× Mexico’s patent intensity but only about 2× its fixed broadband.

TABLE III
WIPO INDICATOR 4A: PATENT PUBLICATIONS BY TECHNOLOGY FIELD,
TOTAL COUNT BY FILING OFFICE (ALL ORIGINS), 2023 [5].

Field	MX	UK	EE	CL	CO
Computer technology	400	1,447	—	66	40
Digital communication	451	775	1	47	78
Medical technology	1,040	645	1	84	158

— = blank in the WIPO file (zero or not reported). 2024 omitted: incomplete.

Fig. 2. Evidence of the WIPO IP Statistics Data Center query.

IX. Materiality, AI and Energy

Physical elements: wearables/phones, mobile networks, servers and data centers. **Material cost:** devices and batteries need silicon, metals and lithium, and must be replaced. **Energy cost:** charging, radio networks and 24/7 server operation and cooling. **Externality:** e-waste and upstream mining (take-back and refurbishment program). **AI helps energy:** ML forecasts of occupancy and weather schedule hospital HVAC to cut peak demand. **AI strains energy:** data centers used about 415 TWh in 2024 (about 1.5% of world electricity) and may reach about 945 TWh by 2030 [9]; continuous inference over wearable streams adds permanent load. We use light tabular models, edge pre-processing and report energy per scored patient.

X. Final Proposal

Problem: late detection of chronic-patient deterioration. **Evidence:** Ensanut Continua 2021–2024 reports 17.0% prevalence of type-2 diabetes [4]; Part I reports 97.4% 4G coverage and 83.1% Internet use [1]–[3], supporting the relevance of chronic care and the feasibility of a mobile channel. These indicators do not establish preventable ED use or clinical effectiveness; both require pilot evaluation. **Strategy:** wearable + app capture, quality control, personal baseline, 48 h risk model, prescriptive outreach, hospital pilot. **Decision:** same-day active contact for the urgent tier instead of a passive visit. **Value:** patients, hospitals, payers, researchers. **Model:** B2B SaaS per active patient. **Limitations:** digital divide and device cost; no clinical validation or regulatory clearance yet; alert fatigue; legacy records; privacy; unproven causal effect; energy and e-waste; Part I indicators do not measure hospital

infrastructure. **Next step:** controlled pilot with pre-registered metrics.

References

- [1] World Bank, “World Development Indicators / Indicators API,” data-helpdesk.worldbank.org.
- [2] ITU, “ITU DataHub,” datahub.itu.int.
- [3] UNCTAD, “UNCTADstat Data Centre,” unctadstat.unctad.org.
- [4] INSP, “Prevalencia de diabetes y control glucémico en México, Ensanut Continua 2021–2024,” *Salud Pública de México*. saludpublica.mx/index.php/spm/article/view/17286
- [5] WIPO, “IP Statistics Data Center,” www3.wipo.int/ipstats
- [6] WIPO, “IPC subclass G16H: Healthcare informatics” (IPC 2018.01). wipo.int/ipc (scheme g16h.pdf)
- [7] WIPO, “IPC and technology concordance table,” IPC/CE/41/5, 2009.
- [8] WIPO, “Frequently Asked Questions: Patents,” sections “What is a patent?” and “What conditions must be met to obtain patent protection?” https://www.wipo.int/en/web/patents/faq_patents
- [9] IEA, “Energy and AI,” 2025. iea.org/reports/energy-and-ai